Nima Kelidari

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Education

University of Southern California - Viterbi School of Engineering Los Angeles, CA

M.S. in Computer Science - Artificial Intelligence | GPA: 3.55/4.0 (3.8/4.0 in AI/ML) Dec 2024 – Dec 2026 (expected)

- Coursework: Machine Learning, Deep Learning, Applied NLP, Multimodal Probabilistic Learning, Analysis of Algorithms, Robot Manipulation
- **Technical Skills – Languages: Python** (Proficient), Java (Proficient), **C/C++**, SQL, CUDA (CuPy) | **ML and LLMs: PyTorch**, JAX, TensorFlow, Hugging Face, **vLLM**, **LangChain/LangGraph**, RAG, MCP, scikit-learn | **Infra: SLURM**, Docker, Kubernetes, FastAPI, Weights & Biases, GCP, AWS, GitHub Actions | **Methods:** evaluation design, interpretability, numerical optimization

Sharif University of Technology Tehran, TE

Bachelor of Science in Computer Science | GPA: 3.37/4.0 Sep 2019 – Feb 2024

- **3.5/4.0 across 15+ CS/AI courses**: Data Structures, Algorithm Analysis, Image Processing, Computer Vision, Deep Learning, Statistical Learning

Experience

University of Southern California - Viterbi School of Engineering Los Angeles, CA

Graduate Student Researcher Aug 2025 – Present

- **Steering frozen LLMs**: designed and trained a **197K-parameter** head that steers MoE routing at inference with zero base-model weight changes, **+0.101 mean faithfulness** over six benchmarks on OLMoE (vs +0.025 for SteerMoE, ICLR 2026) and **+0.288** where retrieved documents conflict with each other, transferring across **six MoE families, 1.3B–30.5B, 16–128 experts**
- **Model interpretability**: trained TopK **sparse autoencoders** (8,192 features, 32 active) on **six speech encoders** across four Parkinson's datasets in three languages (295 speakers), kept only features whose association replicates across datasets, and tested them by intervention: steering along them shifts the classifier's predictions toward Parkinson's in **22 of 22** configurations
- **Agent evaluation**: used a fixed rule-based expert as the yardstick across **100+ controlled runs** to isolate five training choices that lift a PPO/TRPO agent from ∼30% to **34.2 ± 2.1%** against it, and ruled out four, including imitation of an LLM guide and a live LLM opponent; first-author **AIIDE 2026** paper
- **Decision-focused learning**: co-first author of an accepted **NeurIPS 2026 ML×OR workshop** paper that trains outcome models **through Lagrangian dual prices**; in a queueing simulation it stayed **within capacity on all 6 datasets** while the standard pipeline overloaded 4 (up to **116%**), with up to **+72%** policy value

Memorial Sloan Kettering Cancer Center - Department of Medical Physics New York, NY (Remote)

Graduate Research Assistant Sep 2026 – Present

- Benchmarked **Warm-IP**, the group's interior-point QP solver for radiotherapy planning, against its cold-start IP, Gurobi, MOSEK, Clarabel, PIQP and OSQP on **282 public problems**, adding **Bayesian tuning (Optuna TPE)** for all seven and our accelerated ADMM inside four of them

Computational Optimization Intern May 2026 – Sep 2026

- Accelerated the convex QP behind IMRT radiotherapy planning, a path-following ADMM warm-starting an interior-point method on clinical instances up to **380K variables and 574K constraints**, replacing hand-tuning with Bayesian optimization: **78 → 21 iterations** to a usable warm start, 3.7× fewer and **3.2–3.8× less wall-clock** on a held-out clinical case

Sharif University of Technology Tehran, TE

Research Assistant (Computer Vision) and Teaching Assistant Sep 2021 – Aug 2023

- Built **Vision Transformer and 3D CNN generative pipelines** on HPC/GPU clusters in a four-person computer-vision group (Dr. Tefagh); taught **300+ students** across 6 CS courses over 4 semesters

Projects

Footnote: Cited RAG Assistant for SEC Filings (Personal) GitHub

- Built a **RAG assistant** that answers questions about SEC filings and cites its pages: page-bounded chunks, hybrid **BM25 + vector search** with a cross-encoder reranker, routing by company and year, and a **LangGraph** agent with one bounded retry, served through **FastAPI** and an **MCP** server; **85.8%** correct on all 150 FinanceBench questions (87.2% on the 99 held back) vs 33.6% with no documents
- Tested what it answers after a filing changes: with both versions of an amended page in context, answers led with the old amount in up to **100%** of cases, and filing dates plus one instruction sentence brought that to **0% on three models**; an injected instruction was followed in **0 of 630** answers

Tri-Modal Sentiment Analysis, Benchmarked and Deployed (Personal) GitHub

- Built and deployed a text/face/voice sentiment system at **74.5% accuracy / 0.743 weighted-F1 on MELD** (n=2,607), served on **scale-to-zero Cloud Run** behind **FastAPI** with a public webcam-and-microphone playground

Publications

- **A Gold-Standard Study of What Makes a Lightweight Game-Playing Agent Strong**. **First author**. **Accepted, AIIDE 2026 (AAAI)**, Belo Horizonte, Nov 2026. arXiv:2607.06854
- **Differentiating Through Dual Prices: End-to-End Policy Learning Under Capacity Constraints**. **Co-first author** (equal contribution). **Accepted, NeurIPS 2026 ML×OR Workshop**. arXiv:2608.04669
- **Interpreting Speech-Based Parkinson's Disease Detection with Sparse Autoencoders**. **First author**. **Under review, ICASSP 2027**
- **3D-DELTA: A Learned Low-Rank Expert-Steering Head for Frozen Mixture-of-Experts LLMs**. **First author**. Full paper in preparation

Leadership & Activities

- **Ranked 85th nationally (top 0.05%)**, National University Entrance Exam (Math & Physics) | **11th of 1,200 teams**, Pardis Code programming contest | **Top 10 groups**, Mustafa Prize innovation competition
- **B.S. thesis graded 20/20**, Sharif University | **Merit scholarship**, USC Viterbi | **3 merged pull requests** to **PettingZoo** (Farama Foundation) | **CITI Program** certification | **50+ public repositories**